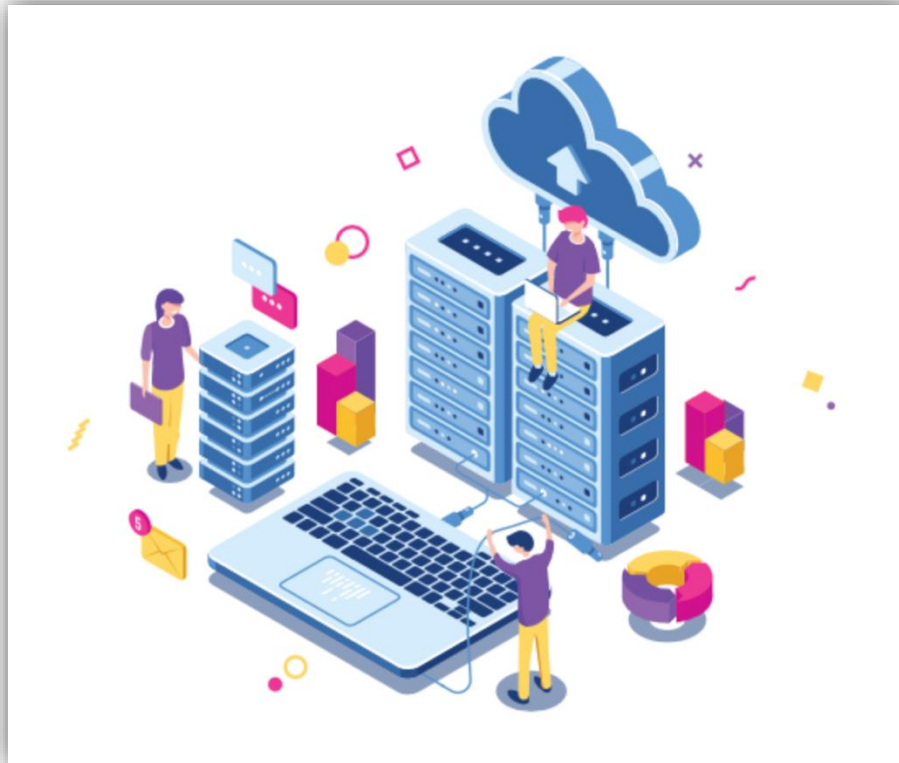




Coles College Faculty and Doctoral Student Alignment Database Solution

Project Database Solution and Documentation - Deliverable 2

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IS 3280 – Data Management

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Executive Summary

The Coles College Faculty and Doctoral Student Alignment Database Solution addresses the need for an efficient and comprehensive system to manage faculty research and teaching interests, align them with the doctoral program, and meet the needs and interests of Ph.D. students. The solution aims to streamline the process of tracking the status and progress of doctoral students throughout their program, ensuring better alignment and coordination between faculty and students.

Problem Description:

Coles College faces significant challenges in maintaining an updated and accessible record of faculty research and teaching interests, aligning these with doctoral program requirements, and tracking the status of Ph.D. students. The absence of a centralized database leads to inefficiencies and difficulties in matching faculty expertise with student needs, ultimately affecting the quality of mentorship and the overall doctoral program experience. Currently, the process relies on disparate systems and manual updates, which are prone to errors and inconsistencies.

Scenario Description:

The proposed database solution will provide a menu-driven, user-friendly interface that allows for easy navigation and manipulation of data. Users will be able to input, search, and update information on faculty and doctoral students, facilitating better alignment of research interests and program requirements. The system will also track the progress of doctoral students, providing valuable insights into their academic journey and milestones achieved.

- **Faculty Management:** Faculty members can enter and update their research interests, teaching subjects, and availability for mentoring doctoral students. This data will be searchable and sortable, enabling quick matches with student interests.
- **Student Management:** Doctoral students can enter their research interests, progress reports, and milestones. Program administrators can track student progress, including coursework completion, research progress, and dissertation status.
- **Alignment Features:** The system will include functionalities to match faculty and student interests, suggest potential mentors, and identify gaps where additional faculty support might be needed.

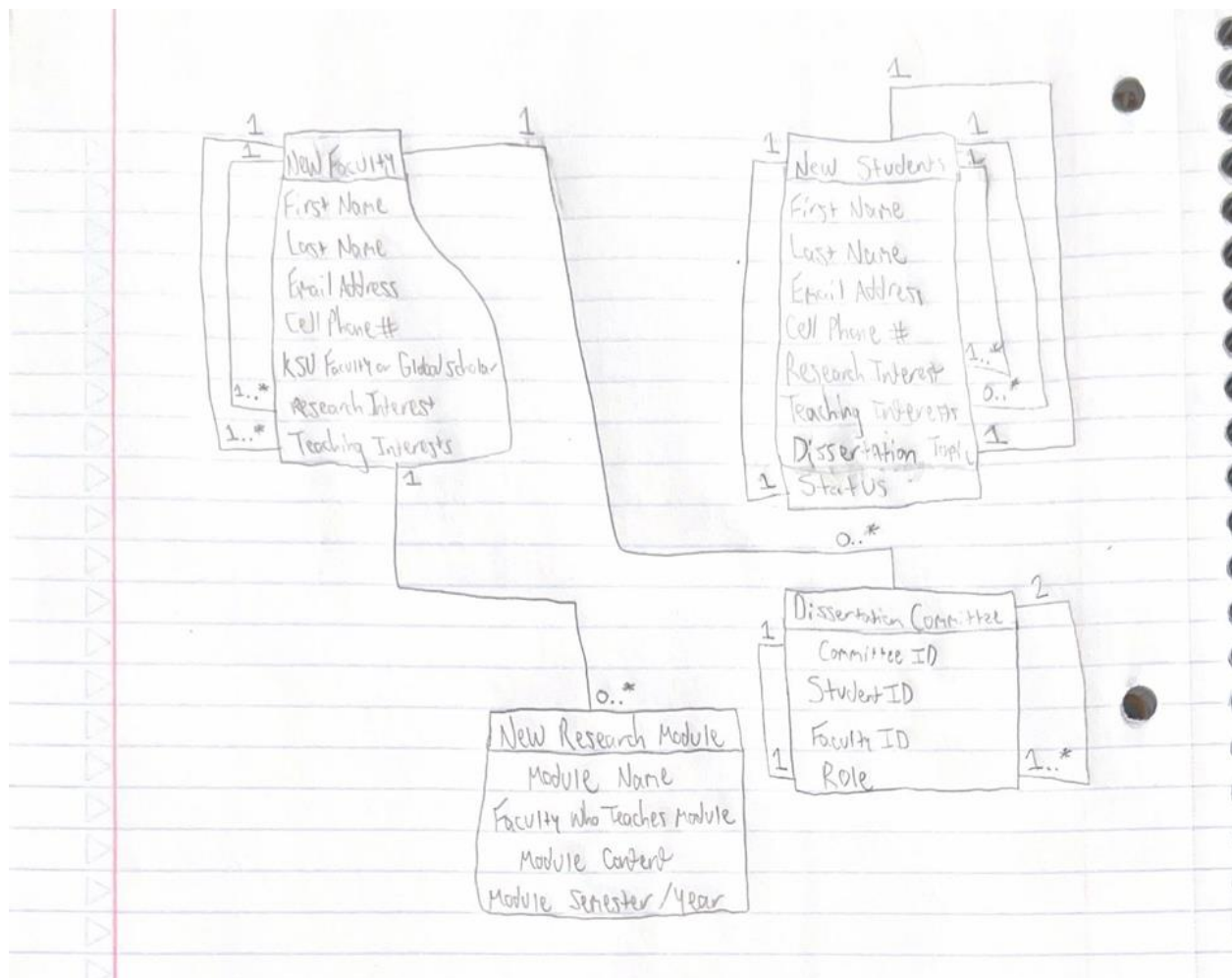
Intended Audience:

- **Sponsor:** Coles College of Business Administration.
- **Primary Users:** Faculty members, doctoral program administrators, and Ph.D. students.

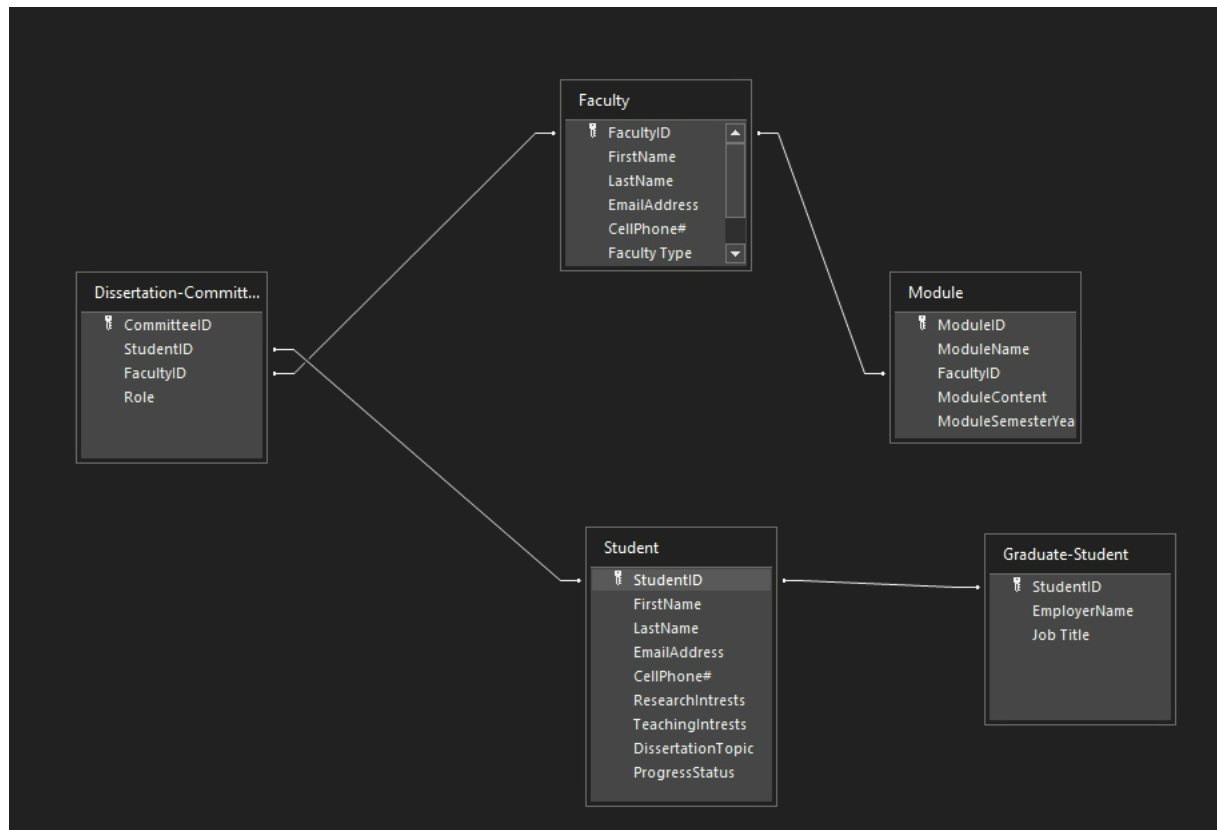
Faculty members will benefit from a streamlined process for managing their mentoring commitments and aligning their research interests with student needs. Doctoral program administrators will have a comprehensive tool for tracking student progress and ensuring program requirements are met. Ph.D. students will have an accessible platform to find mentors whose research interests align with their own and to monitor their academic journey.
- **Secondary Users:** Institutional researchers and academic advisors who may need to generate reports on faculty-student alignment, research outcomes, and program effectiveness. These stakeholders will benefit from the enhanced data integrity and reporting capabilities of the system.

By implementing this database solution, Coles College aims to enhance administrative efficiency, improve the alignment between faculty and doctoral students, and ensure the doctoral program operates smoothly and effectively. The centralized system will contribute to a more organized, transparent, and supportive academic environment, ultimately enhancing the quality of education and research at Coles College.

Class Diagram



Relational Model



Data Dictionary

Table: Programs

Attribute	Data Type	Nullability	Description	Constraints
ProgramID	INT	NOT NULL	Unique identifier for the program	Primary Key
ProgramName	VARCHAR (255)	NOT NULL	Name of doctoral program	
Description	TEXT	NULL	Description of program	

Table: Faculty

Attribute	Data Type	Nullability	Description	Constraints
FacultyID	INT	NULL	Unique identifier for faculty	Primary Key
FirstName	VARCHAR (100)	NULL	Faculty's first name	
LastName	VARCHAR (100)	NULL	Faculty's last name	
ResearchArea	VARCHAR (255)	NULL	Area of research	
TeachingInterest	VARCHAR (255)	NULL	Teaching interests	

Table: Students

Attribute	Data Type	Nullability	Description	Constraints
StudentID	INT	NOT NULL	Unique identifier for student	Primary Key
FirstName	VARCHAR (100)	NOT NULL	Student's first name	
LastName	VARCHAR (100)	NOT NULL	Student's last name	
ResearchInterest	VARCHAR (255)	NOT NULL	Research interests	
Status	VARCHAR (50)	NOT NULL	Current status (e.g., enrolled, graduated)	CHECK (Status IN ('Enrolled', 'Graduated', 'Dropped'))

Table: StudentProgram

Attribute	Data Type	Nullability	Description	Constraints
StudentProgramID	INT	NOT NULL	Unique identifier for the student-program relationship	Primary Key
StudentID	INT	NOT NULL	ID of student	Foreign Key
Program ID	INT	NOT NULL	ID of program	Foreign Key
StartDate	DATE	NOT NULL	Date student started the program	
EndDate	DATE	NULL	Date student ended the program	

Table: FacultyStudent

Attribute	Data Type	Nullability	Description	Constraints
FacultyStudentID	INT	NOT NULL	Unique identifier for the faculty-student relationship	Primary Key
FacultyID	INT	NOT NULL	ID of faculty	Foreign Key
StudentID	INT	NOT NULL	ID of student	Foreign Key
Role	VARCHAR (50)	NOT NULL	Role of the faculty (e.g., advisor, committee member)	

List of Queries

Dissertation Committee, this report generates the list of students and their matching committee members:

```
SELECT Student.StudentID, Student.FirstName + ' ' + Student.LastName AS StudentName,
Faculty.FacultyID, Faculty.FirstName + ' ' + Faculty.LastName AS CommitteeMember,
[Dissertation-Committee].Role
```

```
FROM ([Dissertation-Committee] INNER JOIN Student ON [Dissertation-Committee].StudentID = CStr(Student.StudentID)) INNER JOIN Faculty ON [Dissertation-Committee].FacultyID = CStr(Faculty.FacultyID);
```

Student Numbers, Displays students name and their current employer and title:

```
SELECT [Graduate-Student].StudentID, [Graduate-Student].EmployerName AS EmployerName,  
[Graduate-Student].JobTitle AS JobTitle  
FROM [Graduate-Student];
```

Module, When you enter a module name it displays the professor and info about it.

```
SELECT Faculty.FacultyID, Faculty.FirstName + ' ' + Faculty.LastName AS FacultyName,  
Module.ModuleID, Module.ModuleName AS ModuleName  
FROM [Module] INNER JOIN Faculty ON CInt(Module.FacultyID) = Faculty.FacultyID;
```

Matching Research Interests, pairs students and faculty members based on their common research interests.

```
SELECT Faculty.FacultyID, Faculty.FirstName + ' ' + Faculty.LastName AS FacultyName,  
Faculty.ResearchInt AS FacultyResearchInterest, Student.StudentID, Student.FirstName + ' ' +  
Student.LastName AS StudentName, Student.ResearchInt AS StudentResearchInterest  
FROM Faculty INNER JOIN Student ON Faculty.ResearchInt = Student.ResearchInt;
```

Faculty Interests, displays based on a search faculty members interests.

```
SELECT Faculty.FacultyID, Faculty.FirstName + ' ' + Faculty.LastName AS FacultyName,  
Faculty.TeachingInt AS TeachingInterest  
FROM Faculty;
```

Forms and Reports

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Faculty Table

FacultyID	2
FirstName	Pamila
LastName	Dembla
EmailAddress	pdembla@kennesaw.edu
CellPhone#	(470) 578-2319
Faculty Type	KSU Faculty
ResearchIntrests	SAP
TeachingIntrests	Info Systems

Dissertation Committee Table

CommitteeID	2
StudentID	2
FacultyID	2
Role	

Graduate-Student

StudentID	2
EmployerName	The University of Georgia
Job Title	Instructor of Information systems

Student Table

StudentID	2
FirstName	Michael
LastName	Edwards
EmailAddress	medwards@students.kennesaw.edu
CellPhone#	770-730-2827
ResearchIntrests	SAP
TeachingIntrests	Info Systems
DissertationTopic	SAP use case in non-profit food banks
ProgressStatus	80

Module

ModuleID	1
ModuleName	Literature Review Design for Ed. D. Dissertation
FacultyID	2
ModuleContent	To assist Ed.D in preparing a scientifically rigorous led dissertation.
ModuleSemesterYear	Summer 2024



Matching Research Intrests

Tuesday, July 23, 2024

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FacultyID	FacultyName	FacultyResearchInterest	StudentID	StudentName	StudentResearchInterest
2	Pamila Dembla	info systems	2	Michael Edwards	info systems
3	Saurabh Gupta	info systems	2	Michael Edwards	info systems
4	Aaron French	info systems	2	Michael Edwards	info systems
5	Andy Green	info systems	2	Michael Edwards	info systems
6	Radwan Ali	info systems	2	Michael Edwards	info systems



Faculty Intrest Search

Tuesday, July 23, 2024

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FacultyID	FacultyName	TeachingInterest
2	Pamila Dembla	Info Systems
3	Saurabh Gupta	Info Systems
4	Aaron French	Info Systems



Module Report

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FacultyID	FacultyName	ModuleID	ModuleName
2	Pamila Dembla	1	Literature Review Design for Ed. D. Dissertation



Student Dissertation Commiettee

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StudentID	StudentName	FacultyID	CommitteeMember	Role
2	Michael Edwards	2	Pamila Dembla	Chair
3	Michael Edwards	2	Pamila Dembla	Chair
4	Carolyn Strickland	2	Pamila Dembla	Chair
7	Nicole Wade	3	Saurabh Gupta	Chair
5	Dianna Thompson	4	Aaron French	Chair
6	Brianna Davis	5	Andy Green	Chair
8	Derrick Bejamin	6	Radwan Ali	Chair
7				



Student Jobs

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StudentID	EmployerName	JobTitle
3	The University of Georgia	Instructor of Information systems

Team Reflections:

Khadijah's reflection: My understanding of database technology development greatly increased through this project. I learned a lot about Microsoft Access and was first introduced to it during a course I completed last semester. The knowledge I gained last semester helped me in gaining comfort with Microsoft Access and I was able to complete my table of the project easily. The main challenge I faced during this project was working on Microsoft Access while using VMware. Last semester, I had access to a Windows machine, so I was able to complete any exercises I had to do in Access on that computer. However, this semester I wasn't able to have access to a Windows machine to work on this project, so I had to work on my Mac laptop utilizing VMware. It was my first time working with VMware and it was a bit of a learning curve in order to fully understand it and complete my work in Microsoft Access while using it. There were instances where it was extremely slow or was unable to open up, so it was crucial for me to free up some space on my laptop or close some apps running in the background for VMware to run smoothly. I also ran into an issue where the work that I input into my table in Microsoft Access was not saving to the file everyone in the group had access to. As a group, we had to discuss this issue together and see how we could resolve it. While working on this project, I also had to understand relational models. This semester was the first time I worked with UML during our assignment when we had to work with it. This project deepened my understanding of relational models and how everything needs to be connected within the database. It was also interesting to see Entity-Relationship Diagrams and how every aspect of them relates to one another. It was nice to work on a database and project that will help solve a real-life problem. It is essential for databases to be easy for users to understand and view. The users should be able to revisit the database and process the information in a structured and intuitive way.

Mona's reflection: Creating the Coles College Faculty and Doctoral Student Alignment

Database Solution provided invaluable insights into database technology development within an academic context. Our goal was to unify faculty research management, align it with doctoral program requirements, and streamline student progress tracking. A major challenge was data fragmentation across multiple systems, hindering efficient faculty-student matching and program coordination. We addressed this by rigorously gathering stakeholder requirements—from faculty to administrators and students—to inform our database design. This process ensured our schema could accommodate diverse needs while adhering to third normal form (3NF) principles. Our approach involved translating business rules into an entity-relationship diagram (ERD), validating it through iterative feedback, and implementing it using SQL and Access. SQL queries were crucial for functionalities like matching students with faculty based on research interests and tracking progress through milestones. Team collaboration was key, fostering innovative problem-solving and effective communication amidst technical challenges. Personally, this project enhanced my skills in database design, normalization, and SQL query formulation, highlighting the iterative refinement process crucial for delivering a user-centric solution. Feedback underscored the database's impact on administrative efficiency and faculty-student collaboration. Looking forward, I aim to apply these experiences in future projects to drive continuous improvement and innovation in database development. In conclusion, the Coles College project deepened my understanding of database technology's role in optimizing academic processes, equipping me with skills vital for future endeavors in technology-driven environments.

Nick Bahr's Reflection: This project has taught me a lot about databases and their importance in not only just business. While completing the project, it allowed me to understand relationships between tables in access and their importance when getting important information. Though this project taught me more about databases, there were still many challenges that I and my group faced. This project also allowed me to learn new access features and how we can use SQL in access databases. One of the biggest challenges was just not being extremely familiar with access. Reading the rubric, we saw that our database needs to be menu driven, which at the time none of us knew how to make possible. Like most computer programs out, there are videos all around the web showing you step by step how to do certain tasks. After discovering a short but informative video, I was able to make our access file into a menu driven database. For me, this changed my whole perception of the program. When first introduced to access, I understood that it was a database, and you build relationships based on the different databases. By making it menu driven it felt more like a website instead of a database. Another challenge we faced was working together on the access file. While using Office 365, we were able to open the shared file and download it but, when someone inserted information into tables it would not save. This led us to have to email a file back and forth which made it longer to complete the project. I personally have not investigated this, and I am also not sure if the IT crash currently going on had any play with this. For future reference I will look more into this problem and see if there are ways for multiple people to work on a single access file. Overall, this project brought me a deeper understanding of access and all its capabilities.

Max Harris' Reflection: Creating this solution for the Coles college faculty and doctoral program has taught me a lot of things about database management and Microsoft Access. Working on this project provided me some useful insight into the complexities of existing systems that do this for KSU. It is easy to think something can just be done with data, but someone must go through and build out these systems, searches, queries and reports for the end users. Throughout this project I got to learn more about Microsoft access and SQL, as prior to this project and class I had a very limited knowledge of both. Creating a menu-driven database solution was a big learning curve for me and the entire group, but working together and using resources available to us, like D2L, YouTube and our common knowledge we were able to solve it. The successful completion of this database solution provided me and the entire group a lot of useful knowledge that we will be able to use beyond this class in real work job situations and to assist us during interviews. My biggest personal challenge after figuring out how to build out the tables, was making sure correct data went into them. It is easy to just throw random information into the tables but if you have mismatch information between tables, it causes issues with SQL queries. Verifying firstly the information the group created and found was correct, then entering it carefully before testing my queries was extremely important. It showed me, data is only as good as the source and person entering it, we talked about this in a discussion post recently in class as well. Technology is controlled by people, so it can fail just like people can, it was important to remember this when building this database solution. When thinking about how a user will use it vs how it was designed helped me think of problems and build in solutions before finishing the project helps the long-term usefulness of the solution. Overall, this project taught me a lot about the actual workings of a Microsoft Access database solution, not only the application of it but the ideas and concepts behind it.

Mariah Weldon's Reflection: Throughout the completion of the project focused on maintaining faculty research and teaching interests at Coles College and aligning them with the doctoral program and Ph.D. student needs, I have gained a deeper understanding of database technology development. Database development involves the design, implementation, and management of systems to store, retrieve, and manipulate data efficiently. One of the significant challenges I faced was ensuring data accuracy and consistency. Addressing data redundancy and ensuring referential integrity were critical tasks. Additionally, integrating diverse data types and sources—such as faculty research interests, teaching records, and student status—posed significant difficulties. Creating a comprehensive schema to accommodate various data points seamlessly was essential. Furthermore, ensuring the database's scalability and flexibility was crucial to adapt to future needs and changes. This involved selecting an appropriate database management system and designing efficient queries to handle complex data retrieval. Another challenge I faced was that it was also my first time using the Access DBMS, so I had to gain a thorough understanding of it to effectively operate it. This project emphasizes the importance of a well-designed database in enhancing data management and decision-making processes within an organization. It highlighted that effective database technology development requires a balance between technical expertise and an understanding of user requirements, ultimately leading to a system that is both robust and user-friendly.

